Beta reduction and physical clearance in a parametric booster-to-storage-ring transfer line

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Abstract

Reducing intermediate beta functions is often used to improve transfer-line robustness, but its benefit depends on where apertures, dispersion and trajectory errors limit the beam. We compare two quadrupole settings for a parametric 21.789 m, 4 GeV booster-to-storage-ring transfer line with nine independently adjustable quadrupoles. Both settings reach the same configured exit optics within declared tolerances. Constrained minimization reduces the vertical peak beta from 50.75 m to 28.18 m, yet the nominal minimum five-sigma projected clearance at four model aperture stations remains 17.45 mm because the fixed entrance is limiting. On 200 independent held-out static machines with three independent shots each, the paired mean clearance change is 0.0146 mm with a 95% machine-cluster bootstrap interval of −0.0179 to +0.0483 mm; no benefit is resolved at this precision. Independent hard-edge maps, coupled covariance propagation and particle tracking validate the local model. No losses are observed in 1.2 million tracked particles per setting, with nonzero uncertainty bounds reported. A separate audit retains the historical optical-feasibility statistics without interpreting them as physical loss. The comparison shows why beta reduction alone does not establish a clearance gain. The conclusions remain conditional on configured boundaries, discrete apertures and assumed errors, rather than measured hardware or storage-ring capture.

Keywords: transfer-line optics; beam clearance; quadrupole matching; coupled transport; uncertainty

1 Introduction

Diffraction-limited storage rings require control of the distribution delivered to their injection region [1]. Korea-4GSR provides a relevant 4 GeV design context [2, 3], but a numerically matched exit ellipse is only one part of transport feasibility. An intermediate beta maximum is neither a physical aperture nor a trajectory-error tolerance. Dispersion, centroid displacement, coupling and the location of restrictions also matter.

Published transfer-line studies illustrate this distinction. SLS 2.0 separates quadrupole-scan beam characterization from position/angle control and identifies septum restrictions [4]. ESRF–EBS rematching includes off-axis injection-quadrupole fields and mechanical constraints [5]. Sirius studies distinguish injector stages and incoming beam-size effects [6]. MAX IV pulsed-sextupole injection addresses downstream ring capture [7], a different boundary from the one considered here. These primary studies do not establish the particular target or chamber dimensions of our model.

We ask whether lowering vertical peak beta at a fixed exit match increases five-sigma projected clearance at the declared model apertures, and whether a benefit persists under frozen held-out machine errors and independent shot jitter. The result can be negative: a fixed entrance or

a different limiting observable may prevent beta reduction from improving clearance. We use standard optimization, without algorithmic novelty claims. The BTS sequence, archived matched settings and simulation framework are shared with the companion NKM study. This article ends at the BTS exit; it reports neither nonlinear-kicker transparency nor storage-ring capture.

2 Parametric model and physical observables

2.1 Boundaries, coordinates and source limitations

The Accelerator Toolbox (AT)/pyAT model [8, 9] contains 36 elements over 21.789 m, from marker m1 to marker m4 downstream of sept_ex. It has nine 0.2 m quadrupoles in three triplets, three main bends, an extraction kicker and two model septa (Fig. 1). Local coordinates are $(x, p_x, y, p_y, \delta, ct)$ with positions/path length in m, normalized canonical momenta approximately angles in rad, and $\delta = \Delta p/p_0$ dimensionless. The six design bend angles are $+7.5, +88.5, -111.701, +176, -111.701, +88.5$ mrad; reference geometry is held fixed when fields or particle momenta change.

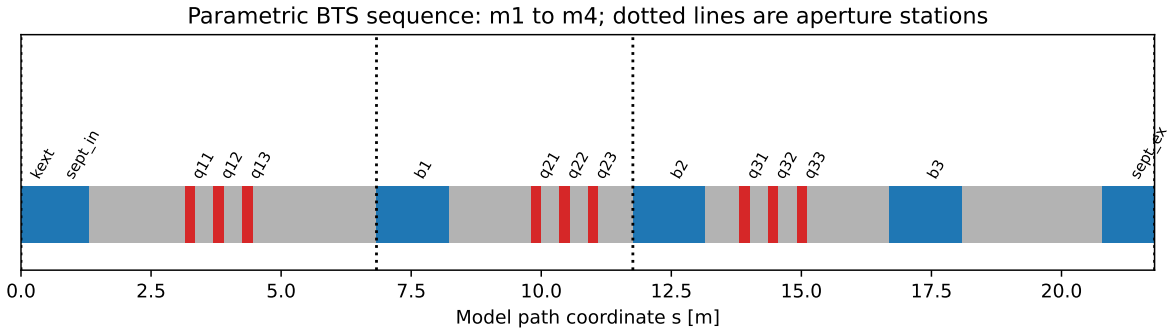


Figure 1: Configured BTS element sequence. Red blocks are quadrupoles, blue blocks are bending elements and grey blocks are drifts. Dotted lines mark the four model aperture stations. The path coordinate follows the reference trajectory; this is not a surveyed mechanical layout.

Table 1: Configured entrance and exit targets. Their physical source at these boundaries is unresolved; the exit is not the NKM-centre ring optics.

Observable	m1 entrance	m4 target
β_x, β_y [m]	7.560000, 12.269000	2.336495, 4.256241
α_x, α_y [1]	1.523100, −1.654700	−0.016335, 0.017772
D_x [m], D'_x [1]	0.276200, −0.065700	0.080868, 0.047472

Entrance and target functions (Table 1) are inherited configured values, treated as parametric. No transport from m4 to a ring matching point is established. Quadrupole family bounds, bore radius, pole-field limit, the 60 m beta screening ceiling and square aperture limits are modelling assumptions, not verified hardware specifications. The legacy MAD-X line has ten active quadrupoles, different drift/bend/edge definitions and an unused additional quadrupole definition. It is not an equivalent independent lattice; our independent validation implements the identical AT sequence analytically.

Aperture elements have half-widths 19.35 mm in both planes at $s = 0$ and 21.789 m, and 30 mm at 6.833 and 11.761 m. The continuous chamber and septum channel restrictions between them are unknown. Station transmission therefore cannot certify machine transmission; no conservative bound for the unknown chamber is claimed.

2.2 Beam covariance and clearance

The new study uses the booster horizontal design input $\epsilon_x = 7.67$ nm rad and $\sigma_\delta = 0.106\%$ [3]. The vertical input $\epsilon_y = 0.767$ nm rad is an assumed 10% ratio. Betatron coordinates and momentum are independent untruncated Gaussians before dispersion is added; initial transverse coupling is zero and $ct = 0$ for this RF-free single pass. The transverse blocks of the complete covariance are

$$\Sigma_{u,\mathrm{bet}} = \epsilon_u \begin{pmatrix} \beta_u & -\alpha_u \\ -\alpha_u & (1+\alpha_u^2)/\beta_u \end{pmatrix}, \qquad \Sigma_0 = \Sigma_{\mathrm{bet}} + \sigma_\delta^2 \mathbf{d}\mathbf{d}^\mathsf{T}, \tag{1}$$

where $\mathbf{d} = (D_x, D_x', D_y, D_y', 1, 0)^\mathsf{T}$. Rolls can generate coupling downstream. Full covariance transport, $\Sigma_s = M_s \Sigma_0 M_s^\mathsf{T}$, retains that coupling and momentum correlation. A local affine map gives the centroid $\bar{\mathbf{z}}_s = M_s \bar{\mathbf{z}}_0 + \mathbf{b}_s$.

For each aperture plane with lower/upper edges a_-, a_+, we define

$$c_u(s) = \min[a_+ - \bar{u}(s), \bar{u}(s) - a_-] - 5\sqrt{\Sigma_{uu}(s)}, \qquad c_{\min} = \min_{s \in \mathrm{stations}, u \in \{x,y\}} c_u(s). \tag{2}$$

Five sigma is a declared projected coverage metric, not complete containment of an unbounded Gaussian. Beta reduction in m must not be called a clearance increase in m. Historical optics-only ensembles use a separate conservative 100/10 nm rad, 0.11% beam configuration and are not pooled with this study.

At 4 GeV the rigidity is approximately 13.34256 T m. For K in m^{-2}, $G = K(B\rho)$ in T/m and $B_{\mathrm{pole}} = |G|r$ in T. Assumed bounds are $|K| \le 2.5$ for q11/q33 and $|K| \le 3$ for the other quadrupoles. With the assumed 19.35 mm bore, the respective maxima are 33.356/40.028 T/m and 0.645/0.775 T, below the assumed 1.2 T screening limit. This calculation checks model consistency, not magnet engineering feasibility.

3 Validation and study design

3.1 Independent maps and active error coverage

We implement exact hard-edge thick-quadrupole and sector-bend maps with edge focusing, translations, roll rotations and fixed-geometry dipole perturbations, without AT optics/matrix calls. They are compared with central differences of AT particle tracking at every element boundary. The declared nominal maximum absolute map tolerance is 2×10^{-6}; transverse symplectic discrepancy must be below 10^{-10}. The original 10-step integrator misses the map criterion (4.05×10^{-6}). Increasing integration to 20, 40 and 80 steps gives 2.53×10^{-7}, 1.58×10^{-8} and 9.89×10^{-10}. New studies use 40 steps; the original miss is retained in the evidence record. Independent exit beta/dispersion agree within 10^{-5} m and 2×10^{-6}, and transverse symplectic discrepancy is 1.22×10^{-15}.

Isolated gradient, translation, roll and dipole-field perturbations pass predeclared local map/orbit checks. Translations drive orbit even when uncoupled Twiss mismatch is insensitive; rolls generate coupling. Additional dipole curvature uses AT PolynomB[0], leaving reference bend geometry fixed. Energy jitter enters particle δ without also rescaling fields. Tracking at $\delta = \pm 0.001$ differs from the nominal linear horizontal prediction by about 0.19 micrometres. For 50,000 design-beam particles, tracked/linear RMS ratios are 0.99896/1.00155, within the declared 2% check. Intentional scraping verifies the first-loss element and surviving-coordinate accounting.

These checks validate a local model. Nonlinear chromatic effects are retained in particle tracking but not the linear clearance estimate; no longitudinal 6D-map symplectic certificate is inferred from a transverse check. Longitudinal magnet placement, supply quantization and downstream NKM/ring errors are outside the new error model. Their absence is not evidence of physical tolerance.

3.2 Matching, alternative design and held-out protocol

Exit residuals for $(\beta_x, \beta_y, \alpha_x, \alpha_y, D_x, D'_x)$ are scaled by $(0.05\text{ m}, 0.05\text{ m}, 0.01, 0.01, 0.002\text{ m}, 0.001)$. The historical search uses bounded least squares and conditional bounded SLSQP, then physical post-selection; it does not impose nonlinear beta constraints during those stages. A replay preserves the nominal start and two random uniform box starts (seed 42, budget 100). Raw and six-decimal settings are independently checked against the stated optical scales, family bounds and positive station clearance.

The alternative minimizes vertical peak beta sampled at element boundaries with six exact exit-residual equalities, horizontal peak beta ≤ 60 m and identical K bounds. SLSQP converges in 67 iterations from the archived match (maximum 200, termination tolerance 10^{-8}). It is a nominal low-beta strategy, not a direct optimizer of held-out clearance. A separate 60-machine training audit (seed 71001) precedes evaluation; no test results select or tune the design. No optical correction or per-shot retuning is performed.

Held-out machines (seed 71002) have independent Gaussian quadrupole relative gradient errors 0.1%, x/y offsets 100 micrometres, rolls 0.5 mrad and dipole relative fields 0.05%. Each of 200 machines has three independent shots with 5% beta fluctuations in both planes (alpha fixed), horizontal centroid RMS 0.5 mm, momentum-angle RMS 0.2 mrad and centroid momentum RMS 0.1%. These are illustrative errors, not sourced engineering specifications. Shot covariance and dispersive centroid terms are included explicitly. Both designs receive the same latent errors and particle draws.

Each shot tracks 2,000 Gaussian particles (seed 71003), yielding 1.2 million per design. Loss is first nonfinite x after an element; surviving coordinates must be finite. All element exits are observed and first-loss indices retained. We estimate paired differences by resampling the 200 machine-average results, with 10,000 percentile-bootstrap replicates (seed 71004). Particles and shots sharing a machine are not treated as independent machines.

4 Results

4.1 Nominal reproduction and local response

The historical three-start replay reproduces one feasible success, one infeasible success and one budget failure (Table 2). The successful infeasible candidate illustrates why matching and feasibility are separate; the failed candidate is not another low-residual match. Three starts do not establish optimizer reliability, search topology or feasible volume.

Table 2: Historical replay outcomes. The initial start is nominal, not random. J is the scaled residual sum of squares. Feasibility is assessed after search.

Start	J	peak β_x [m]	peak β_y [m]	outcome
Nominal	1.29×10^{-13}	31.437	50.746	feasible success
Random 1	1.21×10^{-12}	25.329	62.556	infeasible success
Random 2	1284.86	69.921	65.043	budget failure

The archived raw/rounded strengths remain feasible at 40-step resolution and independent-map precision. Table 3 supplies numerical values; decimal digits do not prescribe power-supply accuracy. Optical functions, dispersion and projected RMS sizes appear in Fig. 2. Beta peaks reported here are element-boundary samples rather than continuous maxima.

For local sensitivity diagnostics we normalize rows by the optical scales and columns by an assumed common 0.01 m^{-2} tuning increment: $A_{ij} = (\partial O_i/\partial K_j)\Delta K_j/\sigma_i$. Central steps $10^{-4}, 10^{-5}, 10^{-6}$ m^{-2} converge; the six active modes and three-dimensional local null space have condition number 57.78074. Six finite small-mode perturbations remain inside bounds and agree

Table 3: Reference and alternative settings [m^{-2}], with reference gradient [T/m] and assumed-bore pole field [T]. Negative K indicates defocusing in x.

Quad	Reference K	Alternative K	Reference G	$\lvert B_{\mathrm{pole}}\rvert$
q11	-0.367525	0.060274	-4.904	0.095
q12	-1.216433	-2.126974	-16.230	0.314
q13	1.586072	1.869755	21.162	0.409
q21	-1.068373	0.954499	-14.255	0.276
q22	2.823108	-0.258649	37.667	0.729
q23	-2.490968	-1.197779	-33.236	0.643
q31	2.042678	2.817163	27.255	0.527
q32	-1.434271	-3.000000	-19.137	0.370
q33	0.351115	0.901619	4.685	0.091

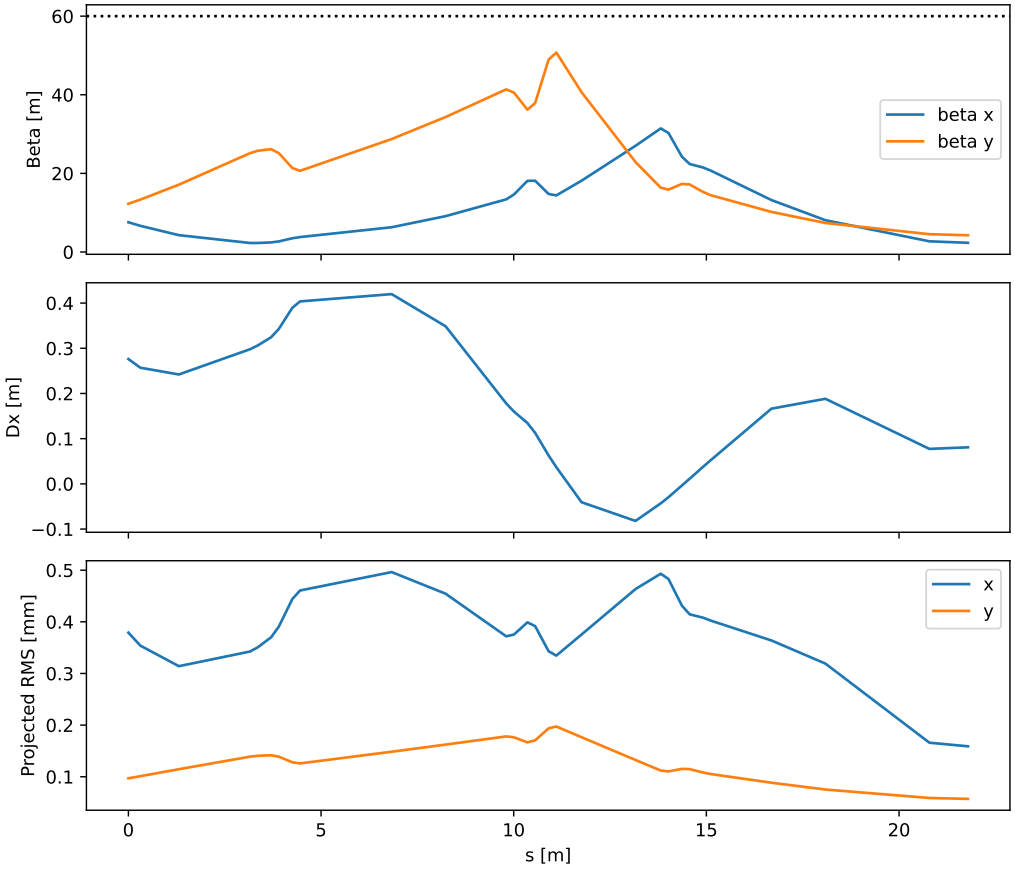


Figure 2: Reference element-boundary beta, horizontal dispersion and projected RMS sizes for the design beam. The dotted beta line is the assumed optical screening ceiling, not a vacuum-chamber edge.

with linear predictions within 1%. This differs from the archived mixed-unit condition number 1421.636. Neither Euclidean singular-vector orthogonality nor a local null space guarantees finite optical decoupling. Achieved commissioning correction claims are withdrawn: measurements, actuator calibration and estimation uncertainty have not been established.

4.2 Beta reduction versus physical clearance

The alternative reduces vertical peak beta from 50.746 to 28.18004 m, while horizontal peak beta changes from 31.437 to 33.57950 m. At zero error, both have the same minimum projected station clearance, 17.45461 mm, set by the fixed entrance. Thus a 22.57 m beta reduction produces no nominal minimum-clearance gain under these apertures.

Held-out mean clearances are 16.86677 mm (reference) and 16.88139 mm (alternative). The paired change is 0.01461 mm, 95% interval [-0.01791, 0.04832] mm (Fig. 3). The interval includes zero, so a clearance benefit is not resolved at this precision. This is not proof of exact equivalence or universal inferiority of low-beta matching. It is a conditional negative result for the selected metric, boundaries, aperture stations and error distributions.

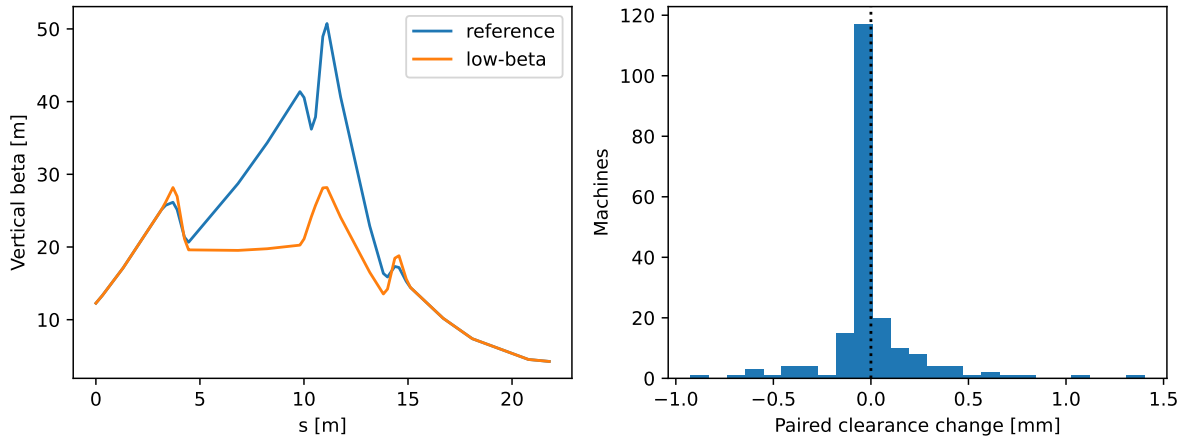


Figure 3: Left: vertical beta reduction at the fixed target. Right: paired changes in machine-average minimum five-sigma station clearance for 200 held-out machines. The bootstrap interval crosses zero despite the beta reduction.

Both designs transmit every tracked particle through the model stations. There are no first-loss entries in this ensemble; a separate deliberate scraping validation verifies loss accounting. Zero observations do not imply zero loss probability. A descriptive particle-binomial calculation gives a two-sided 95% upper loss bound of 3.0741×10^{-6} conditional on an independence assumption. Shared machine/shot effects limit that interpretation. For independent machine blocks with no losses, the one-sided 95% upper probability of a block with any loss is 1.48670% (200 blocks). Observed zero paired transmission change is therefore not quoted as "100% ± 0%" certainty. Neither statement addresses unmodelled continuous restrictions or downstream capture.

4.3 Separate historical statistical audit

Reconstruction of all 550,000 historical observations confirms five optical-rule exceedances, no invalid evaluations, and compatible model/target/error inputs across four runs. The rule is $\beta_{x,\max}$ or $\beta_{y,\max} > 60.01$ m, or $\mathcal{M}_x + \mathcal{M}_y > 2.05$, where $\mathcal{M}_u = \frac{1}{2}\mathrm{Tr}(\Sigma_{u,t}^{-1}\Sigma_{u,o}) - 1$ uses unit-emittance transverse matrices. All five failures exceed vertical beta only (60.23–61.64 m); they are not physical losses.

The descriptive optical-failure estimate is $5/550000 = 9.09 \times 10^{-6}$ with exact asymmetric 95% limits $[2.95 \times 10^{-6}, 2.12 \times 10^{-5}]$; Wilson limits are $[3.8831 \times 10^{-6}, 2.1283 \times 10^{-5}]$ [10]. The

pooled horizontal median 0.00208629 has the order-statistic interval [0.00207403, 0.00209890], not a pooled-bootstrap interval. Reconstructed median, 95th and 99th quantile intervals match the archived analysis.

Three 50,000-sample median-prefix changes (20.830, 11.436 and 10.567)$\times 10^{-6}$ miss the 10^{-5} target; the 400,000 follow-up change 2.395×10^{-6} passes. This is median stability, not rare-tail convergence. The follow-up stopping decision is not archived as predeclared; fixed-N intervals do not certify sequential coverage. Historical one-at-a-time sensitivity is mean absolute change of $\mathcal{M}_x + \mathcal{M}_y$, not J. Seed42 ranks beta-only entrance errors (0.00890374), gradients (0.000861122) and energy (0.000106336). Near-zero alignment/roll mismatch does not certify alignment/coupling tolerances. Historical and new-model data remain separate.

5 Discussion and conclusions

The completed comparison identifies a limitation of beta-only optimization: when a fixed boundary controls the selected physical clearance metric, lowering intermediate beta does not necessarily provide a useful margin. Dispersion and centroid response further prevent converting a beta difference into millimetres of clearance. The useful design decision is to optimize and validate the actual limiting observable rather than assume a low-beta setting is better.

This conclusion is deliberately bounded. The target and continuous chamber are not source-confirmed; four square stations and illustrative independent Gaussian errors are assumptions. Sampling precision cannot compensate for these physical uncertainties. The results do not specify construction tolerances, commissioning feedback or machine reliability. With confirmed chamber geometry, a different limiting station or target, the preferred design could change. A more precise paired interval would require more independent machines, not merely more particles in the same shots. A rare-failure campaign is not required to support the present unresolved-clearance result.

Independent maps, covariance, isolated perturbations, finite particle transport and historical reconstruction provide inspectable checks. Nominal matching, physical station transmission and storage-ring capture remain distinct. A final engineering application requires validated boundary/chamber specifications and measurement/calibration information beyond the present parametric study.

Statements and Declarations

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Author contribution. Chong Shik Park is the corresponding author and is responsible for verifying the scientific interpretation, declarations and final article. Final author approval is pending.

Generative assistance. OpenAI Codex assisted with software changes, numerical execution, literature checking and manuscript drafting. This extends beyond copy editing; the author must validate and approve the final results.

Data and code availability. The prepared referee attachment contains selected raw observations, configurations, code snapshots, environment records, hashes and numerical rebuilding instructions. It has not been publicly deposited; the author must provide the attachment to reviewers. The companion NKM study shares upstream BTS assets, as disclosed above.

A Evidence identity and numerical reproduction

The preserved reviewer report evaluates the early main-branch draft at `bee78914`. The revised asset identity, point-by-point response and artifact manifest are provided in the attachment and `docs/paper_bts/revision_selection.json`. Exact artifact-specific hashes and JSON pointers replace a global generating-commit claim; some historical generating revisions remain unestablished. The new driver supplies validation, matching, normalized-response and frozen-design stages. Historical statistics are independently reconstructed from supplied observations; regenerating those old random realizations is not asserted. All stage outputs use new directories. A fresh isolated LaTeX build supplies this one-column article and records layout checks. These technical checks are not a journal acceptance decision.

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